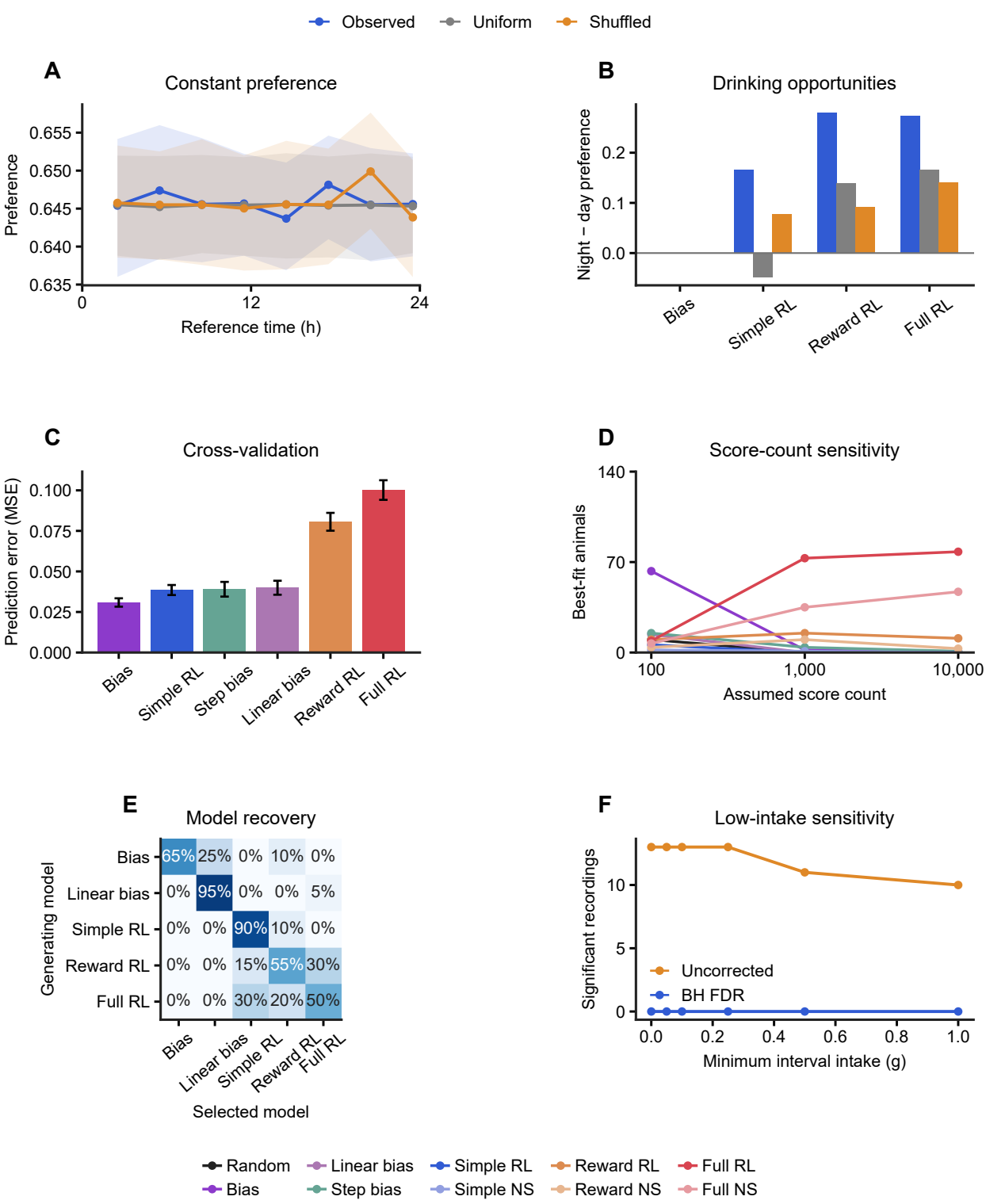

Observed
Uniform
Shuffled
A
Constant preference
Preference
0.655
0.650
0.645
0.640
0.635
0
12
24
Reference time (h)
B
Drinking opportunities
Night − day preference
0.2
0.1
0.0
Bias
Simple RL
Reward RL
Full RL
C
Cross-validation
Prediction error (MSE)
0.100
0.075
0.050
0.025
0.000
Bias
Simple RL
Step bias
Linear bias
Reward RL
Full RL
D
Score-count sensitivity
Best-fit animals
140
70
0
100
1,000
10,000
Assumed score count
E
Model recovery
Generating model
Bias 65% 25% 0% 10% 0%
Linear bias 0% 95% 0% 0% 5%
Simple RL 0% 0% 90% 10% 0%
Reward RL 0% 0% 15% 55% 30%
Full RL 0% 0% 30% 20% 50%
Bias
Linear bias
Simple RL
Reward RL
Full RL
Selected model
F
Low-intake sensitivity
Significant recordings
10
5
0
Uncorrected
BH FDR
0.0
0.2
0.4
0.6
0.8
1.0
Minimum interval intake (g)
Random
Bias
Linear bias
Step bias
Simple RL
Simple NS
Reward RL
Reward NS
Full RL
Full NS